

Physical Chemistry (II)

Lecture 19

Multi-Electron Atoms



Last class

Orbital Approximation

- Each electron's spatial part is represented by an one-electron orbital $\psi_i(\vec{r})$
- Hartree-Fock method provides the “best” orbitals
- Functional forms of orbitals?

Slater-Type Orbitals (STOs)

$$S_{nlm_l}(r, \theta, \phi; \zeta) = \frac{(2\zeta/a_0)^{3/2}}{\sqrt{(2n)!}} \left(\frac{2\zeta r}{a_0} \right)^{n-1} e^{-\zeta r/a_0} Y_l^{m_l}(\theta, \phi)$$

- ζ : orbital exponent
- Linear combination of STOs can be used to model orbitals
 - e.g. double-zeta orbital

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\vec{r}; \zeta_1) + c_2 S_{1s}(\vec{r}; \zeta_2)$$

- Determination of ζ takes most of computational time
 - We can “preset” the values of ζ and optimize

Basis Set

- Atomic orbitals ψ can be represented by linear combinations of some functions ϕ_j

$$\psi(\vec{r}) = \sum_{j=1}^K c_j \phi_j = c_1 \phi_1 + c_2 \phi_2 + \cdots + c_K \phi_K$$

- Basis functions: ϕ_j (e.g. STOs)
- Basis set: $\{\phi_j\}$

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Gaussian-Type Orbitals (GTOs)

$$G_{ijk}(r, \theta, \phi; \alpha) = Nr^{n-1}e^{-\alpha r^2}Y_l^m(\theta, \phi)$$

Computationally efficient

Can be combined to model orbitals



N-MPG Basis Sets

- *N*: number of Gaussian functions for each inner-shell electron orbital
- *M*, *P*: numbers of Gaussian functions for each valence-shell electron orbital

e.g. 3-21G for Lithium (d_i and α_i are preset)

$$\phi_{1s}(\vec{r}) = d_1 G_{1s}(\alpha_1) + d_2 G_{1s}(\alpha_2) + d_3 G_{1s}(\alpha_3)$$

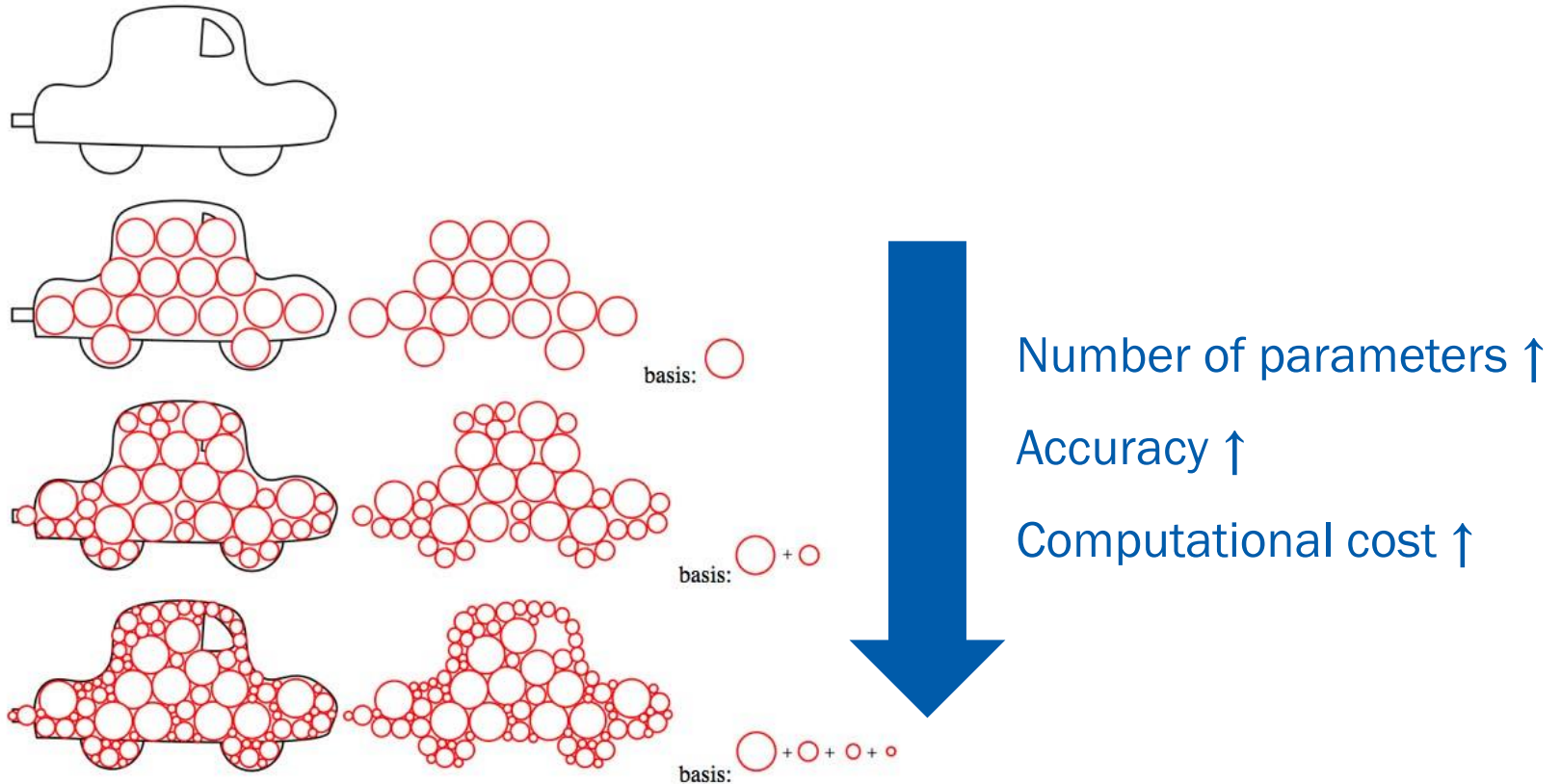
$$\phi_{2s}(\vec{r}) = c_1 \phi_{2s,1}(\vec{r}) + c_2 \phi_{2s,2}(\vec{r})$$

$$\phi_{2s,1}(\vec{r}) = d_4 G_{2s}(\alpha_4) + d_5 G_{2s}(\alpha_5)$$

$$\phi_{2s,2}(\vec{r}) = G_{2s}(\alpha_6)$$

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Size of Basis Set



Extension of HF

- Start from one-electron wavefunctions (“orbitals”) u_i

$$\phi = |u_a \quad u_b \quad \cdots \quad u_n|$$

- HF determinants use **occupied orbitals only**
- **Inclusion of unoccupied (virtual) orbitals** may increase accuracy and take care of correlation

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Example: Virtual Orbital of He

- Double zeta orbital

$$\phi_{1s}(\vec{r}) = c_1 S_{1s}(\vec{r}; \zeta = 1.454) + c_2 S_{1s}(\vec{r}; \zeta = 2.911)$$

- Ground-state orbital (by Hartree-Fock)

$$\psi_0 = 0.844 S_{1s}(\zeta = 1.454) + 0.181 S_{1s}(\zeta = 2.911)$$

- Virtual orbital (orthogonal to ψ_0)

$$\psi_v = 1.624 S_{1s}(\zeta = 1.454) - 1.821 S_{1s}(\zeta = 2.911)$$



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Example: Virtual Orbital of He

- Hartree-Fock wavefunction $\phi_{\text{HF}} = \psi_0(\vec{r}_1)\psi_0(\vec{r}_2)$
- Better wavefunction

$$\phi_{\text{CI}} = c_1\psi_0(\vec{r}_1)\psi_0(\vec{r}_2) + c_2\psi_v(\vec{r}_1)\psi_v(\vec{r}_2)$$

Using a variational method, one can obtain

$$\phi_{\text{CI}} = 0.999 \psi_0(\vec{r}_1)\psi_0(\vec{r}_2) - 0.0476 \psi_v(\vec{r}_1)\psi_v(\vec{r}_2)$$

with optimized energy $E = -2.87541 E_{\text{h}}$

(HF energy $E = -2.86167 E_{\text{h}}$)



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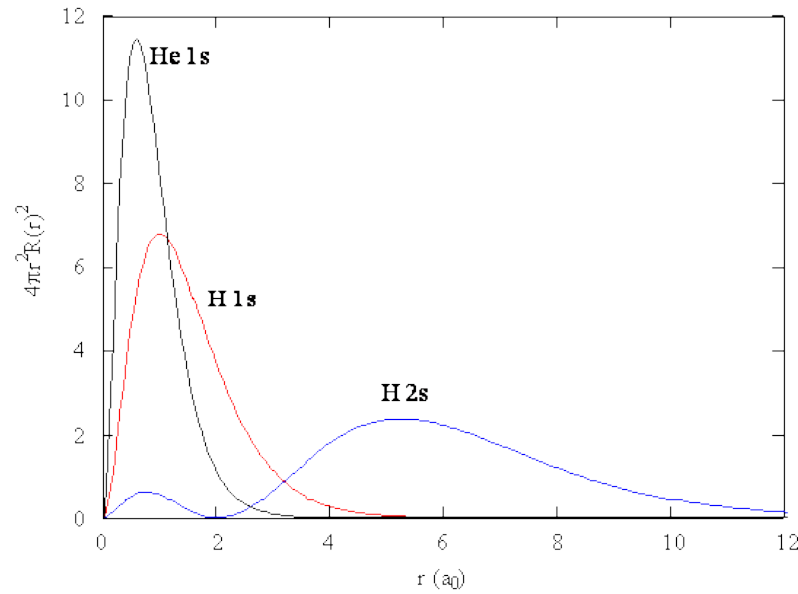
Optimization of Wavefunction

Method		crude $\longrightarrow$ accurate	Applications
Exact solution			H only
Pure VM/PT		depending on complexity	Very few (He)
<i>ab initio</i>	HF		All elements
	CI		All elements
	CC		All elements
	MP		All elements
DFT			All elements

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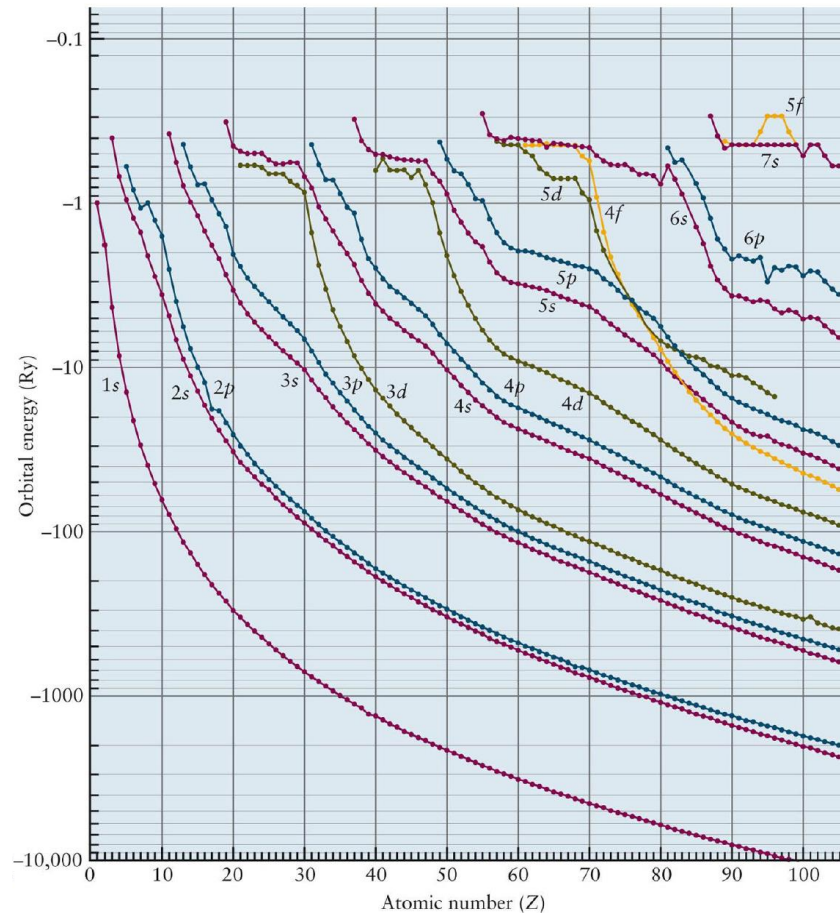
Atomic Orbitals

- **Angular part:** same with H atom (spherical harmonics)
- **Radial part:** slightly different from H atom (but not much)



(Image: <https://chem.libretexts.org/@go/page/4534>)

Orbital Energies (Experimental)



* e-e repulsion leads to *l*-splitting

Electron Configurations

- Distribution of electrons in **atomic orbitals**

e.g. $1s^2$, $1s^1 2s^1$, $1s^2 2s^2 2p^2$

- **Aufbau principle:** ground-state electron configurations

“Electrons first fill orbitals with the lowest available energy”

Electronic States of Atoms

- Ground-state electron configuration of C: $1s^2 2s^2 2p^2$
→ Same configurations can give “different” states
- Angular momentum is a key to the good description
 - For each electron i , we have $\vec{L}_i$ and $\vec{S}_i$ (and $\vec{J}_i = \vec{L}_i + \vec{S}_i$)
- Two possible ways to obtain quantum numbers
 - Russell-Saunders coupling: $\vec{L}_{\text{total}}$ and $\vec{S}_{\text{total}} \rightarrow \vec{J}_{\text{total}}$
 - j-j coupling: $\vec{J}_i$ for each electron $\rightarrow \vec{J}_{\text{total}}$



Russell-Saunders Coupling

- Total orbital angular momentum

$$\vec{L}_{\text{total}} = \sum_i \vec{L}_i$$

- Total spin angular momentum

$$\vec{S}_{\text{total}} = \sum_i \vec{S}_i$$

- Total angular momentum

$$\vec{J}_{\text{total}} = \vec{L}_{\text{total}} + \vec{S}_{\text{total}}$$

Case of Two Electrons

$$\vec{L}_{\text{total}} = \vec{L}_1 + \vec{L}_2, \quad \vec{S}_{\text{total}} = \vec{S}_1 + \vec{S}_2$$

For **nonequivalent electrons** (e.g. $1s^1 2s^1$), the possible quantum numbers are

$$L = (l_1 + l_2), (l_1 + l_2 - 1), \dots, |l_1 - l_2|$$

$$S = (s_1 + s_2), (s_1 + s_2 - 1), \dots, |s_1 - s_2|$$

$$J = (L + S), (L + S - 1), \dots, |L - S|$$

$$M_J = -J, -(J - 1), \dots, J$$

Atomic Term Symbol

$$^{2S+1}[L]_J$$

- Symbols for $[L]$

$L =$	0	1	2	3	4	...
	S	P	D	F	G	

- Closed (sub)shells do not contribute
→ only **open subshells** need to be counted

Two Nonequivalent Electrons

$$1s^1 2s^1: l_1 = 0, l_2 = 0, s_1 = 1/2, s_2 = 1/2$$

$$L = 0; S = 0, 1$$

- $L = 0, S = 0$:

$$J = 0 \rightarrow M_J = 0 (^1S_0)$$

- $L = 0, S = 1$:

$$J = 1 \rightarrow M_J = 0, \pm 1 (^3S_1)$$

Two Nonequivalent Electrons

$$2p^1 3p^1: l_1 = 1, l_2 = 1, s_1 = 1/2, s_2 = 1/2$$

$$L = 0, 1, 2; S = 0, 1$$

- $L = 0, S = 0: J = 0 (^1S_0)$
- $L = 0, S = 1: J = 1 (^3S_1)$
- $L = 1, S = 0: J = 1 (^1P_1)$
- $L = 1, S = 1: J = 0, 1, 2 (^3P_0, ^3P_1, ^3P_2)$
- $L = 2, S = 0: J = 2 (^1D_2)$
- $L = 2, S = 1: J = 1, 2, 3 (^3D_1, ^3D_2, ^3D_3)$

Equivalent Electrons

- **Pauli exclusion principle:** (a, a) impossible
- **Indistinguishability:** (a, b) = (b, a)

1. Construct a table for microstates

$$M_L = m_{l1} + m_{l2} + \cdots, \quad M_S = m_{s1} + m_{s2} + \cdots$$

2. Eliminate the states that violate the two conditions

$$\text{Total number of microstates} = G!/[N!(G - N)!]$$

3. Deduce the possible values of L and S (start from largest M_L and then largest M_S)



Two Equivalent Electrons

$$2p^2: l_1 = 1, l_2 = 1, s_1 = 1/2, s_2 = 1/2$$

$$L = 0, 1, 2; S = 0, 1$$

$$M_L = -2, -1, 0, 1, 2; M_S = -1, 0, 1$$

		M_S		
		1	0	-1
M_L	2	$(1^+, 1^+)$	$(1^+, 1^-)$	$(1^-, 1^-)$
	1	$(0^+, 1^+)$	$(1^+, 0^-); (1^-, 0^+)$	$(0^-, 1^-)$
	0	$(0^+, 0^+)$; $(1^+, -1^+)$	$(1^+, -1^-); (-1^+, 1^-); (0^+, 0^-)$	$(1^-, -1^-); $ $(0^-, 0^-)$
	-1	$(0^+, -1^+)$	$(0^+, -1^-); (0^-, -1^+)$	$(0^-, -1^-)$
	-2	$(-1^+, -1^+)$	$(-1^+, -1^-)$	$(-1^-, -1^-)$

RS 9.9
M 9.10

Allowed Microstates

		M_S		
		1	0	-1
M_L	2		$(1^+, 1^-)$	
	1	$(0^+, 1^+)$	$(1^+, 0^-); (1^-, 0^+)$	$(0^-, 1^-)$
	0	$(1^+, -1^+)$	$(1^+, -1^-); (-1^+, 1^-); (0^+, 0^-)$	$(1^-, -1^-)$
	-1	$(0^+, -1^+)$	$(0^+, -1^-); (0^-, -1^+)$	$(0^-, -1^-)$
	-2		$(-1^+, -1^-)$	

Total number of microstates = $6!/(2!4!) = 15$

RS 9.9
M 9.10

Deducing L and S

		M_S		
		1	0	-1
M_L	2			
	1		 	
	0		  	
	-1		 	
	-2			

- $M_L = 0, \pm 1, \pm 2$ and $M_S = 0 \rightarrow L = 2$ and $S = 0$ (1D)
- $M_L = 0, \pm 1$ and $M_S = 0, \pm 1 \rightarrow L = 1$ and $S = 1$ (3P)
- $M_L = 0$ and $M_S = 0 \rightarrow L = 0$ and $S = 0$ (1S)

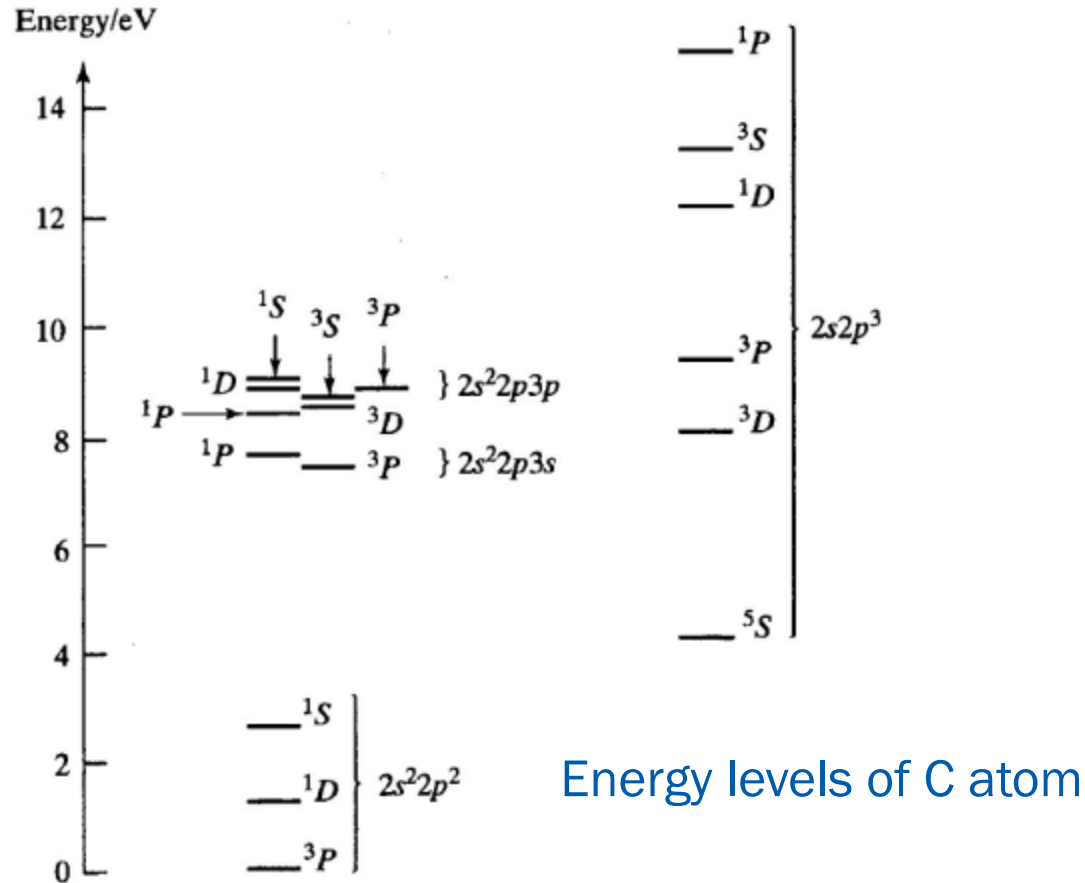
RS 9.9
M 9.10

Configurations and Term Symbols

Configuration	Term symbol
<i>Nonequivalent electrons</i>	
s^1s^1	$^1S, ^3S$
s^1p^1	$^1P, ^3P$
s^1d^1	$^1D, ^3D$
p^1p^1	$^1S, ^3S, ^1P, ^3P, ^1D, ^3D$
<i>Equivalent electrons</i>	
$s^2; p^6; d^{10}$	1S
s^1	2S
$p^1; p^5$	2P
$p^2; p^4$	$^1S, ^1D, ^3P$
p^3	$^2P, ^2D, ^4S$
$d^1; d^9$	2D
$d^2; d^8$	$^1S, ^1D, ^1G, ^3P, ^3F$
$d^3; d^7$	$^2P, ^2D (\times 2), ^2F, ^2G, ^2H, ^4P, ^4F$
$d^4; d^6$	$^1S (\times 2), ^1D (\times 2), ^1F, ^1G (\times 2), ^1I, ^3P (\times 2), ^3D, ^3F (\times 2), ^3G, ^3H, ^5D$
d^5	$^2S, ^2P, ^2D (\times 3), ^2F (\times 2), ^2G (\times 2), ^2H, ^2I, ^4P, ^4D, ^4F, ^4G, ^6S$



Term Symbols and Energies



Energy levels of C atom

Hund's Rules

Empirical rules to find the ground-state term symbol

1. The state with the **largest S** is the most stable
2. For states with the same S , the state with the **largest L** is the most stable
3. For states with the same S and L ,
 1. If a subshell is **less than half-filled**, the state with the **smallest J** is the most stable
 2. If a subshell is **more than half-filled**, the state with the **largest J** is the most stable



Examples

- Beryllium $1s^2 2s^1 3s^1$ (excited state, but ...)

For $s^1 s^1$, we have 3S_1 and 1S_0 as possible term symbols.

Rule 1 $\rightarrow$ 3S_1 is the most stable state.

- Carbon $1s^2 2s^2 2p^2$

For p^2 , we have 1S_0 , 3P_0 , 3P_1 , 3P_2 , 1D_2 .

Rule 1 $\rightarrow$ one of 3P states is the most stable state.

Rule 3 $\rightarrow$ 3P_0 is the most stable state.

Selection Rules

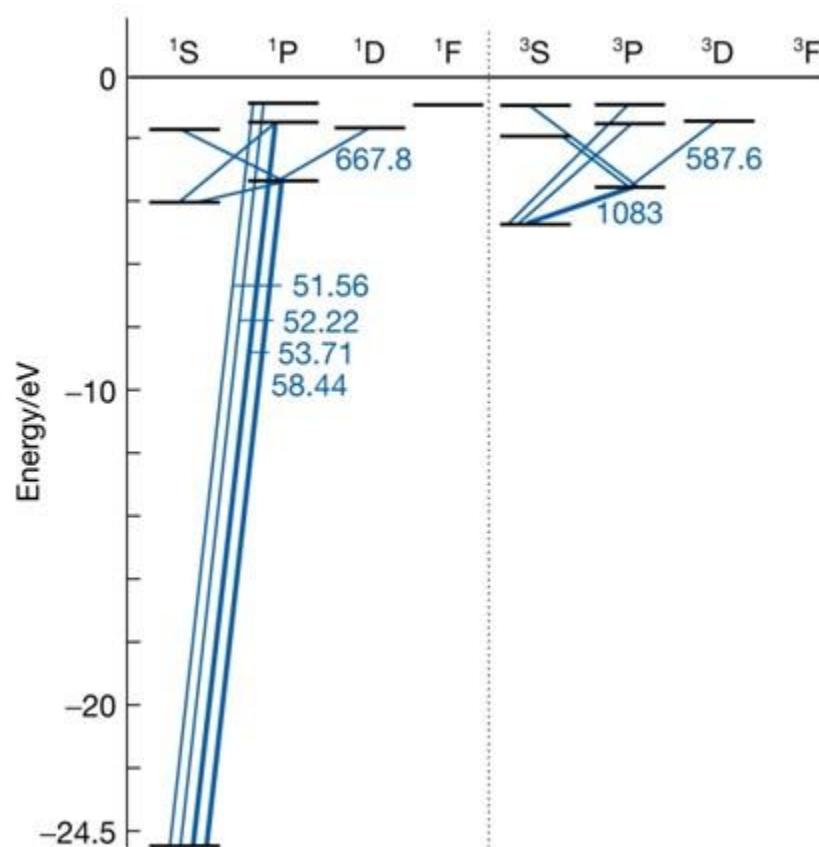
(for multi-electron atoms)

$$\Delta L = 0, \pm 1 \quad (L = 0 \nrightarrow L = 0)$$

$$\Delta S = 0$$

$$\Delta J = 0, \pm 1 \quad (J = 0 \nrightarrow J = 0)$$

Allowed Transitions of Helium



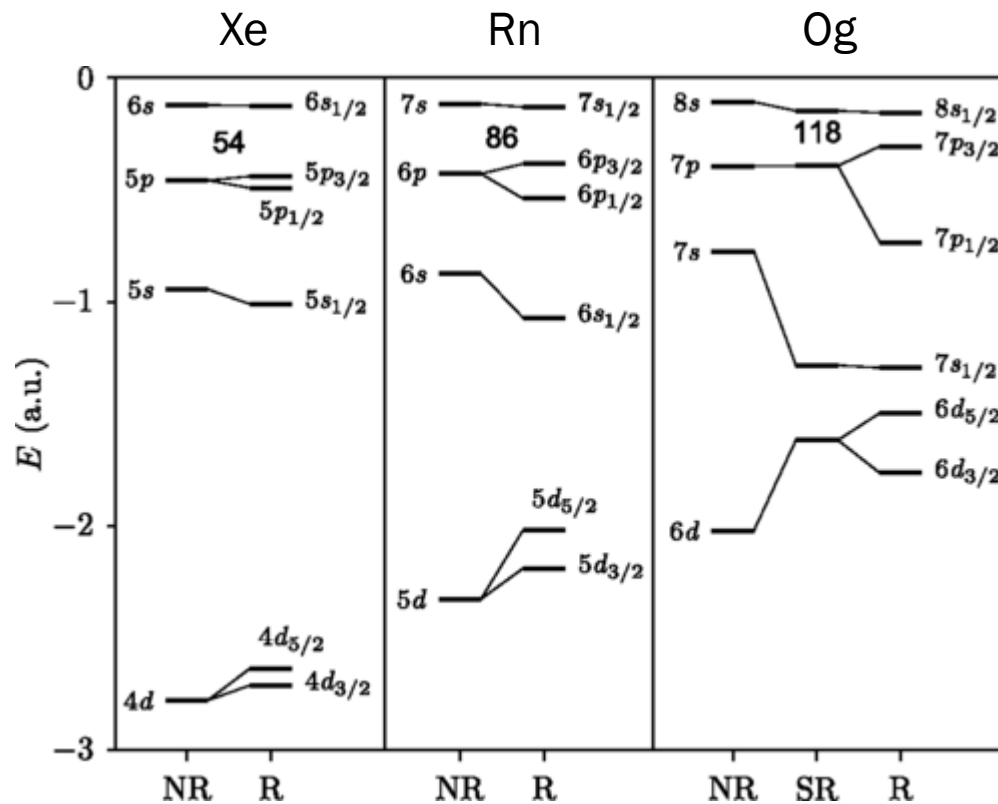
(Image: <https://www1.udel.edu/pchem/C444/spLectures/05062008.pdf>)

Relativistic Effects

- The fine structure of H atom
- The spin-orbit coupling is also often called as relativistic effect
- Both effects become more significant as Z increases
- Larger **spin-orbit coupling**: j - j coupling
- Origin of J dependence in Hund's rule
- Breakdown of the **selection rules**

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Orbital Energies of Inert Gas



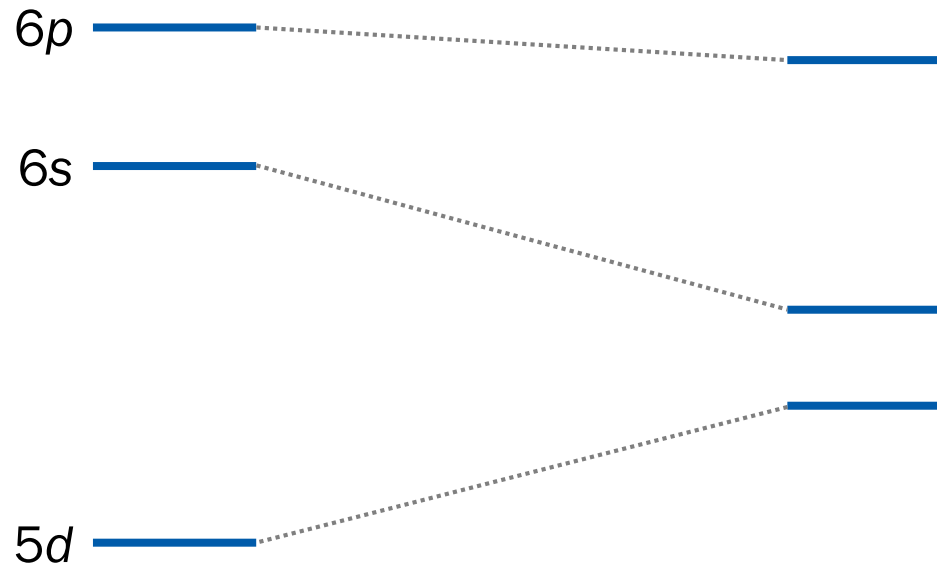
(Image: Jerabek et al., *Phys Rev Lett*, 2018)

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Orbital Energies of Au and Hg

Without relativistic effect

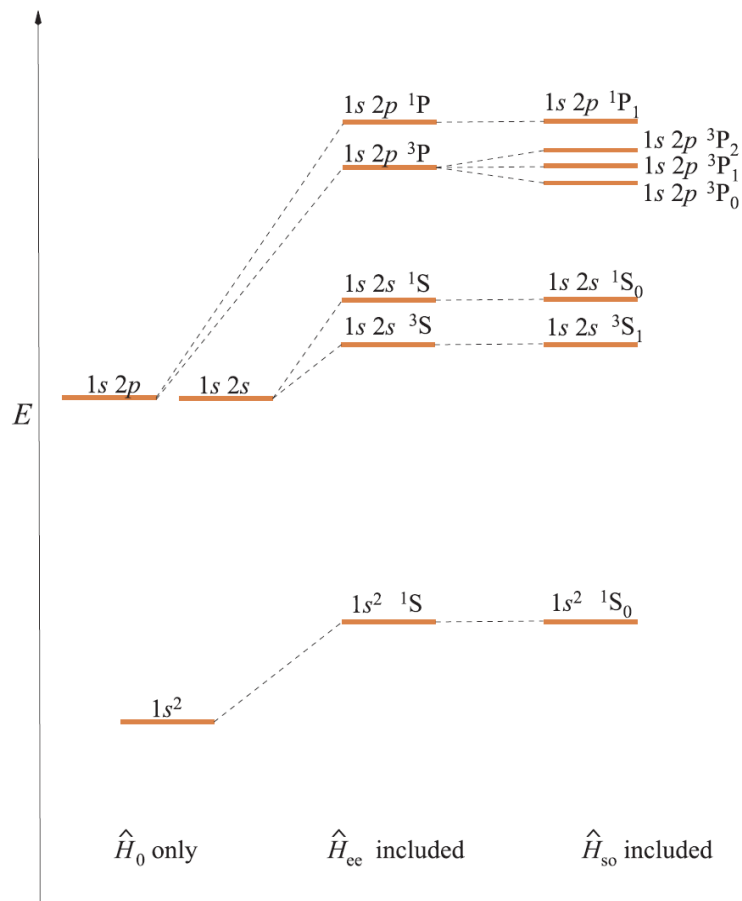
With relativistic effect



- Au: $6s^1 4f^{14} 5d^{10}$
- Hg: $6s^2 4f^{14} 5d^{10}$

RS 9.9
M 9.13

Energy Splitting of Helium



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Fig. 9.11)

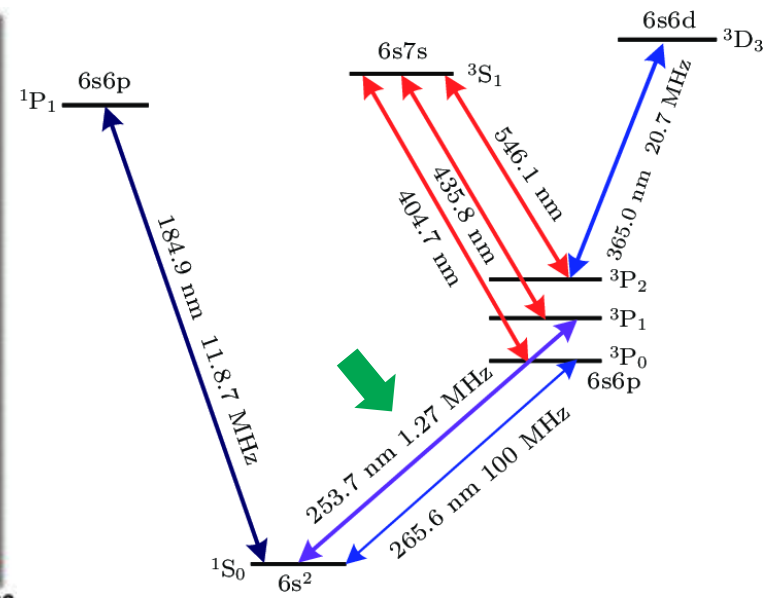
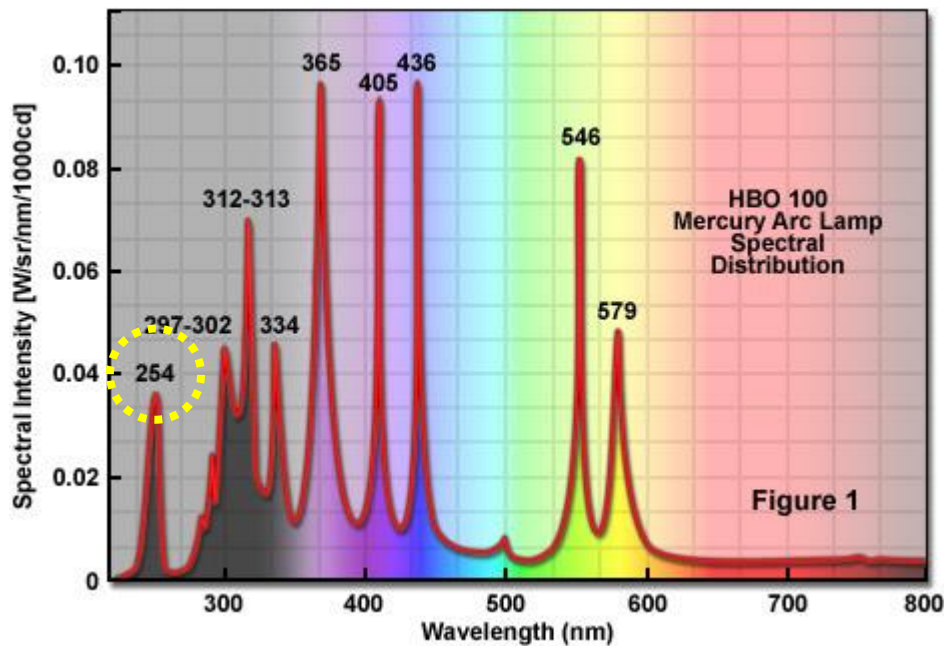
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Mercury Spectrum

$$\Delta L = 0, \pm 1 \quad (L = 0 \rightarrow L = 0)$$

$$\Delta S = 0$$

$$\Delta J = 0, \pm 1 \quad (J = 0 \rightarrow J = 0)$$



(Image: <https://zeiss-campus.magnet.fsu.edu/articles/lightsources/mercuryarc.html>
Liu et al., *Chinese Phys B*, 2013)